

## RHYLL, Australia

WMO: 948920

Lat: 38.4611S

Lon: 145.3100E

Elev: 14

StdP: 101.16

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 6.4        | 7.2 | 1.6                           | 4.2 | 10.4 | 2.6 | 4.6 | 10.3 | 15.3                  | 12.4 | 13.9 | 12.4 | 3.4                   | 0    | 0.519 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 7.3                    | 30.2            | 20.7 | 27.9 | 20.2 | 26.0 | 19.5 | 21.9                | 27.5 | 21.0 | 26.3 | 20.2 | 24.8 | 5.2                  | 350  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.1                            | 14.8 | 24.3 | 19.2 | 14.0 | 23.5 | 18.4 | 13.3 | 22.8 | 64.2          | 27.5 | 61.0 | 26.5 | 58.0 | 24.9 | 26.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 12.5              | 10.9 | 9.4 | 4.6                        | 34.8 | 0.8                | 2.1 | 4.0                                                | 36.3 | 3.5        | 37.5 | 3.0        | 38.6 | 2.4        | 40.1 |    |    |
|                   |      |     | 4.0                        | 24.1 | 0.8                | 1.1 | 3.4                                                | 24.9 | 2.9        | 25.5 | 2.5        | 26.1 | 1.9        | 26.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.0   | 19.4  | 19.8  | 18.4  | 15.9  | 13.5  | 11.4  | 10.7  | 11.2  | 12.5  | 14.1  | 15.9  | 17.5 |
|                                                                                  | DBStd       | 3.95   | 3.14  | 2.95  | 2.88  | 2.31  | 1.75  | 1.49  | 1.33  | 1.58  | 1.95  | 2.45  | 2.88  | 3.16 |
|                                                                                  | HDD10.0     | 21     | 0     | 0     | 0     | 0     | 1     | 4     | 8     | 6     | 2     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 1417   | 22    | 15    | 34    | 79    | 151   | 208   | 238   | 223   | 174   | 133   | 85    | 55   |
|                                                                                  | CDD10.0     | 1846   | 292   | 276   | 261   | 177   | 107   | 47    | 29    | 42    | 77    | 128   | 178   | 233  |
|                                                                                  | CDD18.3     | 201    | 55    | 57    | 37    | 6     | 0     | 0     | 0     | 0     | 0     | 3     | 13    | 30   |
|                                                                                  | CDH23.3     | 1159   | 351   | 312   | 181   | 17    | 0     | 0     | 0     | 0     | 0     | 17    | 76    | 207  |
|                                                                                  | CDH26.7     | 333    | 121   | 89    | 44    | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 64   |
| Wind                                                                             | WSAvg       | 4.3    | 4.4   | 4.3   | 4.2   | 3.9   | 4.0   | 4.2   | 4.7   | 4.4   | 4.3   | 4.2   | 4.3   | 4.3  |
| Precipitation                                                                    | PrecAvg     | 800    | 45    | 52    | 48    | 70    | 74    | 78    | 76    | 80    | 81    | 71    | 67    | 56   |
|                                                                                  | PrecMax     | 996    | 135   | 152   | 79    | 175   | 159   | 148   | 140   | 135   | 150   | 123   | 151   | 120  |
|                                                                                  | PrecMin     | 541    | 5     | 8     | 21    | 22    | 27    | 20    | 36    | 20    | 35    | 17    | 24    | 6    |
|                                                                                  | PrecStd     | 127    | 28    | 33    | 16    | 35    | 33    | 32    | 28    | 31    | 30    | 30    | 30    | 27   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 33.9  | 32.4  | 30.6  | 25.3  | 20.1  | 16.6  | 15.7  | 18.2  | 21.8  | 25.7  | 28.2  | 31.6 |
|                                                                                  |             | MCWB   | 22.3  | 21.1  | 20.4  | 17.5  | 15.0  | 13.0  | 11.6  | 12.9  | 14.8  | 16.9  | 19.7  | 20.6 |
|                                                                                  | 2%          | DB     | 29.9  | 29.3  | 27.6  | 22.9  | 18.5  | 15.4  | 14.5  | 16.4  | 19.2  | 22.7  | 25.5  | 28.3 |
|                                                                                  |             | MCWB   | 20.8  | 21.0  | 19.7  | 16.8  | 14.2  | 12.5  | 11.3  | 12.0  | 13.8  | 16.2  | 18.8  | 20.0 |
|                                                                                  | 5%          | DB     | 27.1  | 27.1  | 25.2  | 21.1  | 17.3  | 14.5  | 13.6  | 15.0  | 17.5  | 20.6  | 23.1  | 25.4 |
|                                                                                  |             | MCWB   | 20.5  | 20.4  | 18.9  | 16.2  | 13.7  | 11.9  | 10.7  | 11.3  | 13.1  | 15.3  | 17.9  | 19.2 |
|                                                                                  | 10%         | DB     | 24.6  | 24.9  | 23.0  | 19.5  | 16.3  | 13.7  | 12.9  | 13.8  | 16.1  | 18.5  | 20.9  | 22.8 |
|                                                                                  |             | MCWB   | 19.7  | 19.7  | 18.3  | 15.6  | 13.3  | 11.5  | 10.4  | 10.7  | 12.4  | 14.5  | 16.8  | 18.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.5  | 22.9  | 22.0  | 18.7  | 15.9  | 14.1  | 12.7  | 13.9  | 15.7  | 18.3  | 21.0  | 21.9 |
|                                                                                  |             | MCDB   | 30.2  | 28.7  | 28.5  | 23.3  | 18.1  | 15.4  | 14.5  | 17.0  | 19.5  | 22.4  | 25.8  | 28.4 |
|                                                                                  | 2%          | WB     | 22.1  | 21.8  | 20.5  | 17.5  | 15.0  | 13.1  | 11.9  | 12.6  | 14.6  | 16.9  | 19.6  | 20.6 |
|                                                                                  |             | MCDB   | 27.3  | 27.1  | 25.5  | 21.1  | 17.3  | 14.5  | 13.5  | 15.2  | 18.1  | 21.4  | 24.0  | 26.3 |
|                                                                                  | 5%          | WB     | 21.0  | 21.0  | 19.5  | 16.7  | 14.3  | 12.4  | 11.2  | 11.7  | 13.6  | 15.8  | 18.3  | 19.6 |
|                                                                                  |             | MCDB   | 26.2  | 25.8  | 23.9  | 19.9  | 16.6  | 14.0  | 12.9  | 14.0  | 16.6  | 19.6  | 22.2  | 24.6 |
|                                                                                  | 10%         | WB     | 19.9  | 20.1  | 18.6  | 16.0  | 13.7  | 11.8  | 10.7  | 11.1  | 12.7  | 14.7  | 17.2  | 18.4 |
|                                                                                  |             | MCDB   | 24.4  | 24.1  | 22.5  | 18.9  | 15.9  | 13.4  | 12.4  | 13.2  | 15.5  | 18.2  | 20.7  | 22.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.4   | 7.3   | 6.7   | 5.7   | 4.5   | 4.1   | 4.2   | 4.9   | 5.8   | 6.5   | 6.8   | 7.2  |
|                                                                                  |             | MCDBR  | 12.7  | 11.8  | 10.9  | 8.6   | 6.1   | 5.0   | 4.9   | 6.5   | 8.1   | 10.4  | 11.2  | 12.7 |
|                                                                                  |             | MCWBR  | 5.9   | 5.4   | 5.0   | 4.2   | 3.4   | 3.2   | 3.0   | 3.6   | 4.4   | 5.5   | 6.0   | 6.3  |
|                                                                                  | 5% WB       | MCDBR  | 11.3  | 10.0  | 9.6   | 7.6   | 5.4   | 4.6   | 4.5   | 5.9   | 7.3   | 9.5   | 10.1  | 11.3 |
|                                                                                  |             | MCWBR  | 5.8   | 5.2   | 4.9   | 4.1   | 3.4   | 3.2   | 3.1   | 3.6   | 4.4   | 5.7   | 5.9   | 6.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.359  | 0.359 | 0.338 | 0.334 | 0.319 | 0.309 | 0.310 | 0.322 | 0.345 | 0.345 | 0.351 | 0.347 |      |
|                                                                                  | taud        | 2.498  | 2.509 | 2.569 | 2.555 | 2.577 | 2.590 | 2.580 | 2.540 | 2.472 | 2.488 | 2.498 | 2.526 |      |
|                                                                                  | Ebn at Noon | 972    | 948   | 929   | 868   | 820   | 799   | 822   | 868   | 905   | 952   | 975   | 990   |      |
|                                                                                  | Edh at Noon | 113    | 107   | 93    | 82    | 70    | 64    | 68    | 82    | 100   | 108   | 112   | 111   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.03   | 6.14  | 4.83  | 3.35  | 2.22  | 1.76  | 1.98  | 2.81  | 4.02  | 5.39  | 6.32  | 7.04  |      |
|                                                                                  | RadStd      | 0.42   | 0.31  | 0.27  | 0.17  | 0.11  | 0.12  | 0.08  | 0.14  | 0.24  | 0.37  | 0.34  | 0.39  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.47 | N/A     | N/A    | +0.80   | N/A   | N/A   | -10         | -144    | +206    | +64     |
| (47) | Regional (32 neighbors) | +0.50 | N/A     | N/A    | +0.78   | N/A   | N/A   | -16         | -134    | +174    | +56     |

Nomenclature: See separate page